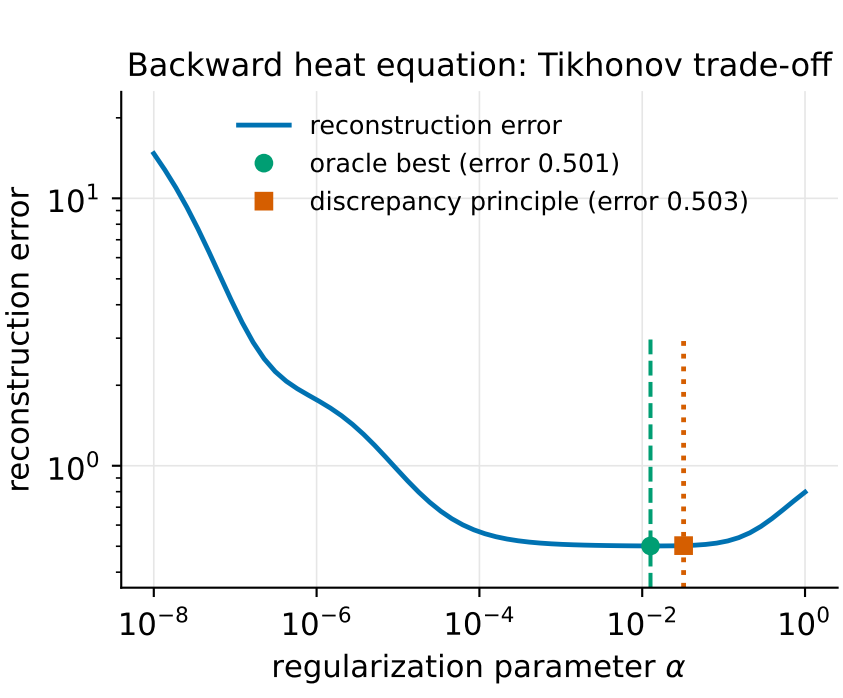
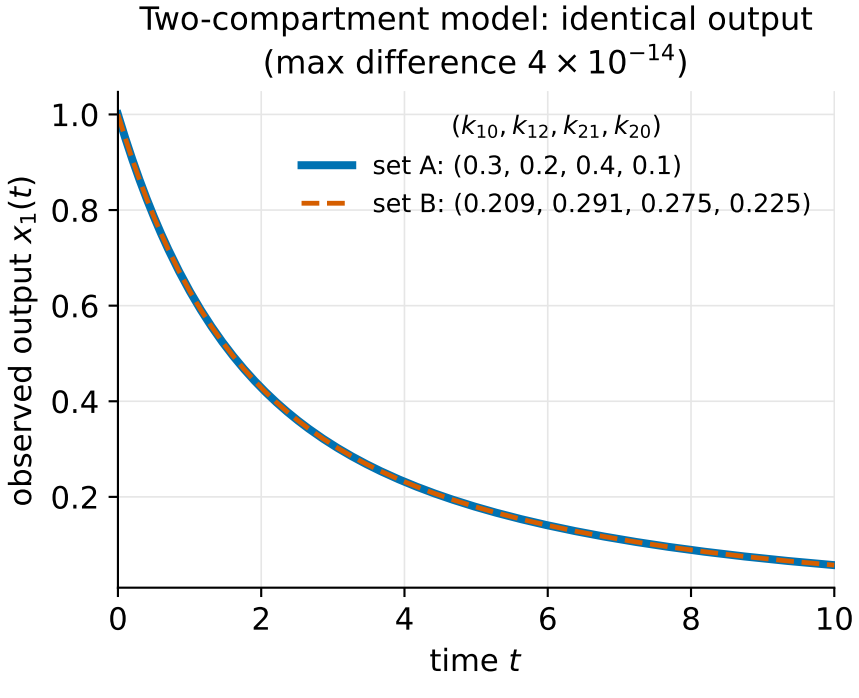

Backward heat equation: Tikhonov trade-off
reconstruction error
oracle best (error 0.501)
discrepancy principle (error 0.503)
reconstruction error
10^1
10^0
10^{-8}
10^{-6}
10^{-4}
10^{-2}
10^0
regularization parameter α


Two-compartment model: identical output
(max difference 4×10^{-14})
$(k_{10}, k_{12}, k_{21}, k_{20})$
set A: (0.3, 0.2, 0.4, 0.1)
set B: (0.209, 0.291, 0.275, 0.225)
observed output $x_1(t)$
1.0
0.8
0.6
0.4
0.2
0
2
4
6
8
10
time t